Bradycardia With a Pulse

AC-02 | Adult Cardiac

Quick flow

01 Assess + Support [EMT / AEMT / Paramedic]

Airway • oxygen for hypoxemia • assist ventilation as needed • monitor pulse, rhythm, SpO_2, and blood pressure • apply pacing/defibrillation pads.

02 Cardiopulmonary Compromise? [EMT / AEMT / Paramedic]

Hypotension • acutely altered mental status • shock • ischemic chest discomfort • acute heart failure. Confirm symptoms are caused by the bradycardia.

03 No Compromise [EMT / AEMT / Paramedic]

Monitor, obtain 12-lead ECG and glucose, identify and treat the cause, obtain 2 complete vital-sign sets, and transport as clinically indicated.

04 Severe Instability / High-Grade Block? [Paramedic]

If severe instability or high-grade AV block is present and vascular access is unavailable, begin transcutaneous pacing immediately while access is pursued.

05 Atropine [Paramedic]

Paramedic: atropine 1 mg IV/IO. Repeat every 3-5 minutes to a maximum total dose of 3 mg. Do not delay pacing when atropine is unlikely to be effective.

06 Persistent Instability [Paramedic]

Transcutaneous pacing and/or epinephrine infusion 2-10 mcg/min or dopamine infusion 5-20 mcg/kg/min; titrate to clinical response.

07 Reassess + Transport [EMT / AEMT / Paramedic]

Confirm mechanical capture, repeat ECG and vital signs, continue cause-directed care, and notify the receiving facility early.

Overview

- Adult bradycardia pathway for patients 16 years of age and older. Bradycardia is typically a heart rate below 50/min, but treatment is determined by the patient's clinical condition rather than the rate alone.
- Treat cardiopulmonary compromise attributable to bradycardia while simultaneously identifying and correcting reversible causes.

EMT Care

BLS stabilization, monitoring, ECG acquisition, pacing-pad placement, and rapid transport.

EMT

- Assess airway, breathing, circulation, mental status, perfusion, and whether the bradycardia is causing the patient's symptoms.
- Provide oxygen for hypoxemia and assist ventilation with BVM when respirations are inadequate.
- Apply cardiac monitor and pacing/defibrillation pads, monitor pulse, blood pressure, and SpO_2, and obtain blood glucose.
- Acquire and transmit a 12-lead ECG when it will not delay urgent treatment or transport.
- Obtain 2 complete vital-sign sets when feasible, assist the ALS provider, and initiate prompt transport for cardiopulmonary compromise or high-grade block.

AEMT Care

All EMT care plus IV/IO access and cautious isotonic fluid support for appropriate hypotension.

AEMT

- Perform all EMT actions and establish IV access without delaying pacing, ALS intercept, or transport.
- For severe instability, make one rapid IV attempt and proceed to IO if unsuccessful or access is otherwise infeasible or delayed.
- For hypotension when volume administration is clinically appropriate, give isotonic fluid in 250-500 mL aliquots with reassessment.
- Use smaller fluid volumes or avoid additional fluid when pulmonary edema, CHF, renal failure, liver failure, or volume overload is suspected.
- Prepare and assist with transcutaneous pacing, medication administration, serial ECGs, and continuous reassessment.

Paramedic Care

All AEMT care plus rhythm interpretation, atropine, transcutaneous pacing, pacing analgesia/sedation, and vasopressor infusions.

Paramedic

- Interpret the rhythm and identify sinus bradycardia, junctional escape, Mobitz I, Mobitz II, complete heart block, or pacemaker malfunction.
- Give atropine 1 mg IV/IO every 3-5 minutes to a maximum total dose of 3 mg when bradycardia is producing cardiopulmonary compromise.
- Do not delay transcutaneous pacing for atropine in severe instability, Mobitz II, complete heart block, or a new wide-QRS escape rhythm.
- If atropine is ineffective, use transcutaneous pacing and/or an epinephrine or dopamine infusion titrated to clinical response.
- Provide analgesia and sedation for pacing when the patient's condition permits, without delaying lifesaving pacing; use continuous cardiac, blood-pressure, respiratory, SpO_2, and $EtCO_2$ monitoring after sedation.
- Confirm and continually reassess mechanical capture, treat reversible causes, and coordinate early receiving-facility notification.

Indications

- Patient 16 years of age or older with clinically significant bradycardia, typically a heart rate below 50/min.
- Bradycardia associated with hypotension, acutely altered mental status, signs of shock, ischemic chest discomfort, or acute heart failure.
- High-grade AV block or another bradyarrhythmia at risk for rapid deterioration.

Contraindications

- Do not provide rate-directed treatment solely for an asymptomatic low heart rate that is appropriate for the clinical condition.
- Do not use this pathway when the patient has no pulse; use the appropriate cardiac-arrest protocol.

Assessment | Clinical Stability

- Determine whether bradycardia is causing hypotension, acutely altered mental status, shock, ischemic chest discomfort, or acute heart failure.
- Assess the relationship between symptom onset and the rhythm; a low heart rate may be normal in healthy adults, athletes, or during sleep.
- Continuously monitor pulse, blood pressure, rhythm, SpO_2, respiratory status, and mental status.

Assessment | Rhythm + Cardiac Evaluation

- Acquire a 12-lead ECG when it will not delay urgent pacing, medication treatment, or transport.
- Identify high-grade AV block, wide-QRS escape rhythms, ischemia/infarction, pacemaker spikes without capture, and rhythm progression.
- Obtain serial ECGs and 2 complete vital-sign sets when feasible, including reassessment after each intervention.

Assessment | Reversible Causes

- Hypoxia, myocardial ischemia or infarction, hyperkalemia or another electrolyte disturbance, acidosis, hypothermia, increased vagal tone, and pacemaker malfunction.
- Medication or toxicologic causes, including beta-blockers, calcium-channel blockers, digoxin, and other rate-slowing agents.

Treatment steps

- Maintain a patent airway, provide oxygen for hypoxemia, assist ventilation as needed, and attach a cardiorespiratory monitor.
- Apply pacing/defibrillation pads early, monitor the pulse continuously, obtain blood glucose, and acquire a 12-lead ECG when it will not delay treatment.
- If no cardiopulmonary compromise is present, monitor and observe, identify and treat the cause, and transport as clinically indicated.
- For cardiopulmonary compromise, establish vascular access while supporting airway, breathing, and perfusion. In severe instability, make one rapid IV attempt and proceed to IO if unsuccessful or delayed.
- Give atropine 1 mg IV/IO every 3-5 minutes to a maximum total dose of 3 mg when appropriate.
- Begin immediate transcutaneous pacing for severe instability or high-grade AV block when vascular access is unavailable. Do not delay pacing for atropine in Mobitz II, complete heart block, or a new wide-QRS escape rhythm.
- For transcutaneous pacing, set a rate of 60-80/min and increase current until electrical capture occurs. Confirm mechanical capture with a palpable pulse, improved blood pressure, pulse-ox waveform, or $EtCO_2$ response - not ECG appearance alone.
- When the patient's condition permits, provide analgesia and sedation using the pain/procedural-sedation protocol; do not delay lifesaving pacing.
- If atropine is ineffective, continue pacing and/or begin epinephrine infusion 2-10 mcg/min or dopamine infusion 5-20 mcg/kg/min, titrated to clinical response.
- Give isotonic fluid only for hypotension when clinically appropriate, using 250-500 mL aliquots with reassessment and reduced volumes for patients at risk of overload.
- Identify and treat reversible causes and use the applicable ACS, CHF, hypothermia, or toxicology pathway when identified.
- Begin transport after immediate stabilization measures, continue serial ECGs and vital signs, and notify the receiving facility early.

Medications

Atropine

1 mg IV/IO; repeat every 3-5 minutes to a maximum total dose of 3 mg

- Paramedic standing order for bradycardia causing cardiopulmonary compromise.
- Do not delay pacing when atropine is unlikely to be effective or the patient is severely unstable.

Epinephrine Infusion

2-10 mcg/min IV/IO infusion; titrate to clinical response

- Paramedic standing order for persistent unstable bradycardia when atropine is ineffective.
- Push-dose epinephrine is not included in this protocol.

Dopamine Infusion

5-20 mcg/kg/min IV/IO infusion; titrate to clinical response

- Paramedic standing order for persistent unstable bradycardia when atropine is ineffective.
- Taper slowly after clinical stabilization when appropriate.

Isotonic Crystalloid

250-500 mL IV/IO aliquots for hypotension when clinically appropriate; reassess after each aliquot

- Use smaller volumes or avoid additional fluid in pulmonary edema, CHF, renal failure, liver failure, or suspected volume overload.

Warnings / precautions

- Treat the patient, not the monitor or a heart-rate number. Asymptomatic bradycardia generally does not require rate-directed treatment.
- Do not delay transcutaneous pacing in severe instability, Mobitz II, complete heart block, or a new wide-QRS escape rhythm.
- Electrical pacing artifacts do not prove perfusion. Confirm mechanical capture and reassess it continuously.
- Transcutaneous pacing is painful in a conscious patient; provide analgesia and sedation when feasible, but never delay lifesaving pacing.
- Do not use push-dose epinephrine under this protocol.
- Avoid routine fluid loading in acute heart failure, pulmonary edema, renal failure, liver failure, or suspected volume overload.

Clinical pearls

- Correction of hypoxia, hyperkalemia, hypothermia, ischemia, or toxicologic causes may be more important than simply increasing the heart rate.
- In severe instability without vascular access, pacing can begin while another clinician pursues IV or IO access.
- Mechanical capture is supported by a pulse corresponding to paced complexes, improved blood pressure or perfusion, a pulse-ox waveform, or an $EtCO_2$ response.
- Prepare for clinical deterioration and cardiac arrest whenever high-grade AV block, a wide escape rhythm, or recurrent loss of capture is present.

Special populations | Acute Coronary Syndrome

- Obtain serial 12-lead ECGs, avoid delaying stabilization, and use AC-04 when ischemia or infarction is suspected.

Special populations | Medication or Toxicologic Bradycardia

- Use TE-07 and contact Poison Control for suspected beta-blocker, calcium-channel blocker, digoxin, or other toxic exposure; antidotal treatment may be required in addition to pacing and pressor support.

Special populations | Hypothermia

- Use TE-05 because medication response, pacing response, handling, and transport priorities may differ in significant hypothermia.

Related pathways / contacts

AC-04 Acute Coronary Syndrome / STEMI - Use when myocardial ischemia or infarction is suspected.

AC-05 CHF / Acute Pulmonary Edema - Use when acute heart failure or pulmonary edema is present.

TE-05 Hypothermia / Frostbite - Use for significant hypothermia-associated bradycardia.

TE-07 Overdose / Toxic Ingestion - Use for medication- or toxin-associated bradycardia.

Source and references

American Heart Association. 2025 Adult Bradycardia With a Pulse Algorithm. | American Heart Association. 2025 Guidelines for CPR and ECC: Adult Advanced Life Support, Initial Management of Bradycardia. | American Heart Association. ACLS Provider Manual Change Notice, updated February 6, 2026. | Tennessee Emergency Medical Services Board. Tennessee EMS Protocol Guidelines, September 2025. | Claiborne EMS medication formulary and related cardiac, toxicology, environmental, pain, and sedation protocols.

Revision: October 10, 2026 | Last verified: October 10, 2026